

## CS360 Nano Quiz - 25.06.02(Mon)

### 1. Please choose a **WRONG** explanation about histogram.

- (1) There are three major types for histogram: equal-width, equal-height, and most-frequent-values.
- (2) In equal-height histogram, the widths of the ranges may be different with each other.
- (3) In equal-width histogram, the heights of the ranges are the same.
- (4) In most-frequent-values histogram for R.a, if the number of the most-frequent-values are seven in the histogram, and  $V(R,a)-100$ , then, the size of others group is 93.
- (5) Using histograms typically provides a more precise estimation of the cost (i.e., the number of tuples) compared to using simple statistics like  $V(R,a)$ .

### 2. Please choose a **WRONG** explanation about the number of possible join trees.

- (1) There are two approaches to exploring the space of possible plans: two-down and bottom-up.
- (2) The dynamic programming method belongs to the bottom-up approach.
- (3) Join trees selects the relation of a smaller estimated size as the left argument conventionally.
- (4) There are three shapes of join trees: left-deep, bushy, and right-deep.
- (5) When the number of relations is n, only a single left-deep join tree is possible.

### 3. Please choose a **WRONG** explanation about dynamic programming for query plan enumeration.

- (1) Reducing the search space is one of the reasons why left-deep join trees are conventionally used.
- (2) The left-deep join tree is usually better than the right-deep join tree, in terms of memory usage and the number of scans.
- (3) In dynamic programming (DP) table entries for join-order optimization, the cost represents the size of intermediate relations.
- (4) In the DP table, the cost is set to 0 when the number of relations is one or two (i.e., the BASIS case).
- (5) Bush join trees always yield the best plan among the three shapes of join trees.

## CS360 Nano Quiz - 25.06.04(Wed)

### 1. Please choose a **WRONG** explanation about completing physical query plan.

- (1) After selecting a join order, we must decide whether intermediate results are materialized or pipelined.
- (2) Selection operation is typically done in two steps: retrieving all tuples satisfying a comparison, and then applying a filter operator to evaluate the rest of selection conditions.
- (3) When choosing a join algorithm, we first attempt to use a one-pass algorithm, hoping that # of available buffers is sufficient.
- (4) Sort-join is always a good choice even when no input relations are sorted on join attribute (s).
- (5) Index-join is a good choice for  $R(a,b) \bowtie S(b,c)$  when R is expected to be small, and an index exists on S.b.

### 2. Please choose a **WRONG** explanation about the hash-based join example.

- (1) Pass 1 partitions both R and S to M-1 buckets using the join attribute as the hash key.
- (2) Pass 2 builds a hash table for a smaller chunk and probes the tuples of a larger chunk.
- (3) We use the number of tuples to estimate the cost of plan, but use the number of blocks (e.g., k) to determine the physical operations within the plan.
- (4) If k is sufficiently small, we can use pipelining; however, if k is large enough, materialization is required.
- (5) No matter how large k is, we can perform a hash-based join for  $(R(w,x) \bowtie S(x,y)) \bowtie U(y,z)$  using  $M = 101$  buffers.

### 3. Please choose a **WRONG** explanation about notation for physical query plans.

- (1) Every DBMS uses the same internal notation.
- (2) Physical operators for leaf nodes include TableScan(R), SortScan(R,L), IndexScan(R,C), and IndexScan(R,A).
- (3) IndexScan(R,A) behaves like TableScan(R), but may be more efficient, if R is not clustered.
- (4) The notation for physical operators at non-leaf nodes typically includes the operation, its parameters strategy, # of passes, and estimated # of memory buffers required.
- (5) The total estimated number of memory buffers required by all physical operators in a plan must remain below a certain threshold.